

Emergency Dispatch Priority and Coordination System

Project Report

ENGE707 Data Engineering and AI

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Table of Content

1. Problem Definition and Dataset Selection.....	3
1.1 Problem Statement.....	3
1.2 Dataset Selection.....	4
2. Data Acquisition and Understanding.....	5
2.1 Data Acquisition and Dataset Structure.....	5
2.2 Initial Data Quality Assessment.....	6
2.3 Data Interpretation and Assumptions.....	6
3. Data Cleansing and Transformation.....	7
3.1 Data Cleaning and Preparation.....	7
3.2 Outlier Investigation and Treatment.....	8
3.3 Cleaned Dataset and Validation.....	8
4. Exploratory Data Analysis.....	9
4.1 Descriptive and Distribution Analysis.....	9
4.2 Relationships and Patterns in the Data.....	10
4.3 Key Findings and Dataset Implications.....	10
5. Data Engineering Pipeline and Phase I Summary.....	11
5.1 Data Engineering Pipeline.....	11
5.2 Phase I Summary and Next Steps.....	11
References.....	12

1. Problem Definition and Dataset Selection

1.1 Problem Statement

There are countless factors in people's daily lives which can influence their health, health decisions such as sleep, physical activity, diet, alcohol intake and other lifestyle habits. Being able to understand how these factors could influence serious health conditions could help people make better lifestyle choices in regards to their health. Heart disease is the biggest global health problem, causing 19.8 million deaths in 2022 alone. Being able to identify lifestyle factors that are associated with heart disease could help people make more cautious health choices, before heart disease develops.

Currently, identification of heart disease requires a major cost as in New Zealand, it approximately cost 538.8 million NZD (Telfar Barnard, 2025). Unaffordability is one of the main causes as well as geographically, and the ability to receive frequent testing for the general population. Even though the primary affected group is the general population, the health care facilities and personnel are also affected. This is because if the government would like to encourage more people to get regular health checks, this would overwhelm the facility thus causing less possibility for individuals that actually need testing. Figuring out the lifestyle that tends to give a higher chance of heart disease by the use of a model would increase its availability to determine whether further actions are needed.

The problem investigated in this project is whether the information of patients such as vitals and lifestyle choices could be used to predict whether or not they have heart disease. Instead of one characteristic, this project will investigate how multiple patient characteristics relate to heart disease and determine which of these variables are useful for accurately predicting heart disease.

The analytical goal of this project is to develop a machine learning model that can predict whether a person belongs to the heart disease category using only demographic and lifestyle variables. This would benefit the stakeholders of our project, these being the **general population, healthcare workers, and researchers.**

1.2 Dataset Selection

The dataset we selected for this project is the “*281K Patient Dataset for Multi-Disease Prediction*” sourced from Kaggle on 18 August 2026 and is released under the CCO Public Domain licence.

<https://www.kaggle.com/datasets/danishjmeo/281k-patient-dataset-for-multi-disease-prediction?resource=download>

This dataset contains 280,985 patient records and 39 columns, which include clinically recorded vitals, lifestyle choices, demographic information, and documented diseases. This makes the dataset suitable for our problem statement as it contains all these measurements which can be analysed to form a possible prediction of potential heart disease.

This integrated dataset is a combination of 3 Kaggle datasets that are available publicly. The 3 datasets cover diabetes, heart disease, and hypertension. This source credits the 3 datasets mentioned

(<https://www.kaggle.com/datasets/iammustafatz/diabetes-prediction-dataset>,

[kaggle.com/datasets/oktayrdeki/heart-disease](https://www.kaggle.com/datasets/oktayrdeki/heart-disease),

[kaggle.com/datasets/ankushpanday1/hypertension-risk-prediction-dataset](https://www.kaggle.com/datasets/ankushpanday1/hypertension-risk-prediction-dataset))

2. Data Acquisition and Understanding

2.1 Data Acquisition and Dataset Structure

Explain how the dataset was obtained and loaded using Python, then introduce its overall structure.

Include:

- ☐ file/source format
- ☐ Python libraries/tools used
- ☐ number of records and features
- ☐ target variable if applicable
- ☐ numerical/categorical variables
- ☐ data types
- ☐ variable meanings

```
print("Number of records:", df_original.shape[0])
print("Number of columns:", df_original.shape[1])
```

✓ 0.0s

```
Number of records: 280985
Number of columns: 39
```

```
df_original.info()
✓ 0.5s

<class 'pandas.DataFrame'>
RangeIndex: 280985 entries, 0 to 280984
Data columns (total 39 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   composite_key                        280985 non-null  str
1   age_level                           280985 non-null  str
2   gender                              280985 non-null  str
3   bmi_level                           280985 non-null  str
4   smoking                             280985 non-null  str
5   diabetes                            280985 non-null  str
6   age                                  280985 non-null  int64
7   age_normalized                       280985 non-null  float64
8   bmi                                  280985 non-null  float64
9   hypertension                         280985 non-null  float64
10  heart_disease                       280985 non-null  float64
11  HbA1c_level                         280985 non-null  float64
12  glucose                             280985 non-null  float64
13  cholesterol                         280985 non-null  float64
14  sleep_hours                         280985 non-null  float64
15  triglycerides                       280985 non-null  float64
16  physical_activity                   280985 non-null  str
17  family_history                     280985 non-null  str
18  stress_level                       280985 non-null  str
19  low_hdl_cholesterol                 280985 non-null  str
...
37  sublabel                           280985 non-null  str
38  label                              280985 non-null  str
dtypes: float64(19), int64(1), str(19)
memory usage: 83.6 MB
```

In the dataset, there are three different data types: 19 features are in decimals (float64), 1 is an integer (int64), and 19 are strings (str).

Feature	type	Meaning

[illegible]

2.2 Initial Data Quality Inspection

Document what was found during inspection of the raw dataset. Include:

- ☐ missing values
- ☐ duplicate records
- ☐ unusual or invalid values
- ☐ inconsistent categories/formats
- ☐ unexpected data types
- ☐ unique values and value ranges
- ☐ potential outliers
- ☐ any other notable quality problems

```
# Missing values
missing_values = df_original.isnull().sum()
missing_values[missing_values > 0]
```

✓ 0.4s

Series([], dtype: int64)

```
# Duplicate rows
print("Duplicate rows:", df_original.duplicated().sum())
```

✓ 1.7s

Duplicate rows: 2284

```
# Unique values in each column
for column in df_original.columns:
    print(f"{column}: {df_original[column].nunique()} unique values")
```

✓ 0.8s

```
composite_key: 192 unique values
age_level: 4 unique values
gender: 2 unique values
bmi_level: 4 unique values
smoking: 3 unique values
diabetes: 2 unique values
age: 88 unique values
age_normalized: 151 unique values
bmi: 14122 unique values
hypertension: 137 unique values
heart_disease: 125 unique values
HbA1c_level: 192 unique values
glucose: 154 unique values
cholesterol: 323 unique values
sleep_hours: 10092 unique values
triglycerides: 525 unique values
physical_activity: 3 unique values
family_history: 2 unique values
stress_level: 3 unique values
low_hdl_cholesterol: 2 unique values
high_ldl_cholesterol: 2 unique values
blood_pressure: 197 unique values
high_blood_pressure: 2 unique values
sugar_consumption: 3 unique values
crp_level: 9992 unique values
...
source_dataset: 3 unique values
disease_flags: 8 unique values
sublabel: 8 unique values
label: 2 unique values
```

Output is truncated. View as a [scrollable element](#) or open in a [text editor](#). Adjust cell output [settings](#)...

```
# Numerical ranges and summary statistics
df_original.describe().T
```

✓ 0.6s

	count	mean	std	min	25%	50%	75%	max
age	280985.0	49.665865	21.682350	2.000000	32.000000	50.000000	67.000000	89.000000
age_normalized	280985.0	0.547883	0.249222	0.000000	0.344827	0.551724	0.747126	1.000000
bmi	280985.0	27.490663	7.040373	10.010000	22.100000	27.320000	32.700000	95.690000
hypertension	280985.0	0.128218	0.187346	0.000000	0.000000	0.071429	0.217063	1.000000
heart_disease	280985.0	0.073390	0.138243	0.000000	0.000000	0.007299	0.115053	1.000000
HbA1c_level	280985.0	5.947475	0.954072	3.500000	5.384805	5.800000	6.865766	9.000000
glucose	280985.0	135.258646	38.477865	70.000000	100.000000	138.000000	160.000000	300.000000
cholesterol	280985.0	224.045525	35.525530	150.000000	202.000000	225.000000	243.000000	300.000000
sleep_hours	280985.0	6.954652	1.426412	4.000000	6.200000	6.975185	7.700000	10.000000
triglycerides	280985.0	178.995744	67.336085	50.000000	129.000000	179.000000	242.000000	400.000000
blood_pressure	280985.0	149.887200	4.148644	120.000000	148.946429	149.713320	150.720942	180.000000
crp_level	280985.0	7.489127	1.013898	0.003647	7.294863	7.458483	7.758562	14.997087
homocysteine_level	280985.0	12.465326	0.983375	5.000236	12.287763	12.488407	12.686175	19.999037
systolic_bp	280985.0	134.405105	20.542140	90.000000	126.000000	134.167413	143.000000	179.000000
diastolic_bp	280985.0	89.476720	13.667219	60.000000	84.000000	89.504587	95.000000	119.000000
alcohol_intake	280985.0	15.003336	6.830255	0.000000	12.000000	14.982818	17.900000	30.000000
salt_intake	280985.0	8.480577	2.960646	2.000000	7.200000	8.477600	9.800000	15.000000
heart_rate	280985.0	74.497343	11.402996	50.000000	70.000000	74.487247	79.000000	99.000000
hdl	280985.0	64.609147	15.938525	30.000000	58.000000	64.650655	72.000000	99.000000
ldl	280985.0	129.725919	27.352521	70.000000	118.000000	129.983051	141.000000	189.000000

```
# Categorical values
categorical_columns = df_original.select_dtypes(include="object").columns

for column in categorical_columns:
    print(f"\n{column}")
    print(df_original[column].value_counts(dropna=False))
```

✓ 0.4s

Python

C:\Users\leoca\AppData\Local\Temp\ipykernel_10032\1085246056.py:2: Pandas4Warning: For backward compatibility, 'str' dtypes are included by select_dtypes when 'object' dtype is specified. See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-select-dtypes for details on how to write code that works with pandas 2 and 3.

```
categorical_columns = df_original.select_dtypes(include="object").columns
```

```
composite_key
composite_key
Middle_Female_Overweight_Never_No    6097
Mature_Female_Overweight_Never_No    5778
Middle_Female_Obese_Never_No         5020
Mature_Female_Obese_Never_No         4986
Middle_Female_Normal_Never_No        4806
...
Young_Female_Underweight_Current_No  184
Young_Female_Underweight_Never_Yes   176
Young_Male_Underweight_Former_Yes     175
Young_Male_Underweight_Current_Yes    161
Young_Female_Underweight_Former_Yes   147
Name: count, Length: 192, dtype: int64
```

```
age_level
age_level
Mature      87632
Middle      85344
Senior      70225
Young       37784
Name: count, dtype: int64
```

```
gender
...
label
Abnormal    174101
Normal      106884
Name: count, dtype: int64
```

Output is truncated. View as a [scrollable element](#) or open in a [text editor](#). Adjust cell output [settings](#).

```
> ✓ 0.0s
print(df_original["heart_disease"].value_counts(dropna=False))
print("Unique values:", df_original["heart_disease"].unique())

15] ✓ 0.0s

"" heart_disease
0.000000    121656
1.000000     3923
0.222222     3845
0.032258     3074
0.080000     3040
...
0.000776      255
0.218220      240
0.000300      199
0.090909      185
0.166667      169
Name: count, Length: 125, dtype: int64
Unique values: [0.00000000e+00 1.00000000e+00 5.53359684e-02 1.67968750e-01
5.68993953e-02 3.58422939e-03 2.33333333e-01 8.00915332e-03
3.33333333e-01 4.28571429e-02 8.81971466e-02 1.30434783e-01
3.61010830e-03 3.12500000e-01 2.07373272e-02 6.48379052e-02
1.11607143e-01 7.80031201e-03 8.63930886e-02 1.70309395e-03
5.66146736e-02 2.23684211e-01 1.45985401e-02 6.96864111e-03
3.49180768e-03 2.22222222e-01 1.38613861e-01 9.61538462e-02
1.03092784e-02 2.72727273e-01 2.01816347e-03 2.85714286e-01
1.92592593e-01 1.39534884e-01 1.68032787e-01 1.09027475e-02
1.17647059e-01 1.85185185e-01 7.14285714e-02 3.07017544e-02
2.77777778e-02 1.59090909e-02 1.82504720e-02 8.00000000e-02
1.53846154e-01 9.12698413e-02 9.56146657e-02 1.51020408e-01
...
1.75675676e-01 2.50000000e-01 2.17391304e-01 2.18220339e-01
6.52173913e-02 2.99940012e-04 1.40271493e-01 2.06896552e-02
2.37288136e-01 5.80832527e-03 2.34064786e-01 2.16931217e-01
1.66666667e-01]
```

2.3 Data Interpretation and Assumptions

Explain any assumptions required to understand or prepare the dataset, particularly where the dataset documentation is unclear or values require interpretation.

3. Data Cleansing and Transformation

3.1 Data Cleaning and Preparation

Explain the actual cleaning performed and justify the decisions. Include where relevant:

- ☐ missing-value treatment
- ☐ duplicate removal
- ☐ correction of invalid values
- ☐ category standardisation
- ☐ formatting corrections
- ☐ data-type conversion
- ☐ unit conversion
- ☐ reshaping or aggregation

3.2 Outlier Investigation and Treatment

Explain how potential outliers were investigated and whether they were retained, removed, capped, or transformed.

The decision should be justified rather than automatically treating every extreme value as an error.

3.3 Cleaned Dataset and Validation

Show that the cleaning process produced a valid dataset. Summarise:

- ☐ resulting dataset dimensions
- ☐ remaining missing values
- ☐ duplicate count
- ☐ corrected data types/ranges
- ☐ validation checks performed
- ☐ important differences between the raw and cleaned datasets

4. Exploratory Data Analysis

4.1 Descriptive and Distribution Analysis

Analyse the overall characteristics of the important variables. Include relevant:

- ☐ descriptive statistics
- ☐ numerical distributions
- ☐ categorical frequencies
- ☐ histograms
- ☐ boxplots
- ☐ bar charts
- ☐ target/class distribution
- ☐ class imbalance

4.2 Relationships and Patterns in the Data

Investigate relationships between variables and with the target. Include relevant:

- ☐ scatter plots
- ☐ grouped comparisons
- ☐ correlation analysis
- ☐ correlation heatmap
- ☐ relationships with the target
- ☐ missingness patterns
- ☐ unusual patterns or clusters

4.3 Key Findings and Dataset Implications

Bring the EDA together rather than ending with a collection of graphs. Summarise the most important findings about:

- ☐ dataset structure
- ☐ data quality
- ☐ important relationships
- ☐ imbalance
- ☐ unusual patterns
- ☐ apparent usefulness of the dataset for the intended problem
- ☐ issues that may need to be addressed later during modelling

5. Data Engineering Pipeline and Phase I Summary

5.1 Data Engineering Pipeline

Present the required pipeline diagram and briefly explain the overall process:

Raw Dataset → Acquisition → Inspection → Cleaning → Transformation → Validation → Cleaned Dataset → Exploratory Data Analysis

5.2 Phase I Summary and Next Steps

Briefly summarise what was achieved during Phase I and identify what the project will move into during Phase II, such as feature engineering and machine learning.

Heart Disease Target Note

- Heart disease is a broad category, not one specific condition. It can include coronary artery disease, heart failure, arrhythmias, valve disease, and cardiomyopathy.
- In this dataset, heart_disease is only True/False.
- The dataset does not identify the specific type of heart disease.
- Therefore, the project should ask: *Can clinical and lifestyle measurements be used to predict whether a patient has heart disease?*
- Limitation: different heart conditions are grouped into one binary target.

References

1. Cardiovascular diseases are the leading cause of death globally (World Health Organization, 2025).
2. Telfar Barnard, L. (2025, July). Heart disease in Aotearoa: Morbidity, mortality, and service delivery. Prepared for Kia Manawanui Trust, University of Otago.